

Exp. No. 10	Installation and Configuration of Firewall Using iptables
Date:	

AIM:

ALGORITHM:

PROCEDURE:

Open Ubuntu/WSL

Open Command Prompt and type:
wsl
You will enter the Linux terminal.

OUTPUT:

jvars@LAPTOP-S46QB92F:/mnt/c/Users/jvars\$

Install iptables

Update the package repository:

sudo apt update

Install iptables: sudo apt install iptables -y

Check the installed version:

sudo iptables --version

OUTPUT:

```
jvars@LAPTOP-S46QB92F:/mnt/c/Users/jvars$ sudo iptables --version
iptables v1.8.11 (nf_tables)
```

View Existing Firewall Rules

Execute: sudo iptables -L -v -n

OUTPUT:

```
jvars@LAPTOP-S46QB92F:/mnt/c/Users/jvars$ sudo iptables -L -v -n
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source         destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source         destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source         destination
```

Flush Existing Firewall Rules

Execute: sudo iptables -F

Then: sudo iptables -X

Allow Loopback Traffic

Execute: sudo iptables -A INPUT -i lo -j ACCEPT

Allow Established and Related Connections

Execute:

sudo iptables -A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT

Allow SSH Traffic – Port 22

Execute:

```
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

Allow HTTP Traffic – Port 80

Execute: `sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT`

Configure Test Port 8080 – ACCEPT

Execute: `sudo iptables -A INPUT -p tcp --dport 8080 -j ACCEPT`

Configure Test Port 9090 – DROP

Execute: `sudo iptables -A INPUT -p tcp --dport 9090 -j DROP`

Set the Default Firewall Policies

Execute:

```
sudo iptables -P INPUT DROP
```

```
sudo iptables -P FORWARD DROP
```

```
sudo iptables -P OUTPUT ACCEPT
```

Output:

Normally, no output is displayed if the commands execute successfully.

Existing firewall rules are removed, providing a clean environment for the experiment.

Verify the Firewall Rules

Execute: `sudo iptables -L -v -n`

OUTPUT:

```
Chain INPUT (policy DROP 1 packets, 72 bytes)
pkts bytes target    prot opt in     out     source           destination
  0      0 ACCEPT    all  --  lo     *       0.0.0.0/0        0.0.0.0/0
  5 1624 ACCEPT    all  --  *      *       0.0.0.0/0        0.0.0.0/0
  0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0        0.0.0.0/0
  0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0        0.0.0.0/0
  0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0        0.0.0.0/0
  0      0 DROP      tcp  --  *      *       0.0.0.0/0        0.0.0.0/0
                                     ctstate RELATED,ESTABLISHED
                                     tcp dpt:22
                                     tcp dpt:80
                                     tcp dpt:8080
                                     tcp dpt:9090

Chain FORWARD (policy DROP 0 packets, 0 bytes)
pkts bytes target    prot opt in     out     source           destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target    prot opt in     out     source           destination
```

Test Port 8080

To make the ACCEPT rule observable, first start a simple test service on port 8080:

```
python3 -m http.server 8080
```

OUTPUT:

```
jvars@LAPTOP-S46QB92F:/mnt/c/Users/jvars$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [08/Sep/2026 10:55:06] "GET / HTTP/1.1" 200 -
```

Test Port 9090

A firewall test needs a service listening on the port; otherwise a failed connection could simply mean “no service is running.”

For a clear demonstration, start another test service:

```
python3 -m http.server 9090
```

Final Verification

Execute:

```
sudo iptables -L INPUT -v -n
```

OUTPUT:

```
jvars@LAPTOP-S46QB92F:/mnt/c/Users/jvars$ sudo iptables -L INPUT -v -n
[sudo: authenticate] Password:
Chain INPUT (policy DROP 11 packets, 792 bytes)
pkts bytes target prot opt in out source destination
12 4596 ACCEPT all -- lo * * 0.0.0.0/0 0.0.0.0/0
8 2408 ACCEPT all -- * * 0.0.0.0/0 0.0.0.0/0 ctstate RELATED,ESTABLISHED
0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:22
0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:80
0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:8080
0 0 DROP tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:9090
```

RESULT: